

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CE)

November 2013

CE302.01 System Programming

Date: 22.11.2013, Friday

Time: 10:00 a.m. To 1:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) What is an assembler? What kinds of statements are present in an assembly language program? [05]
- (b) Programming Language (PL) domain bridges the gap between application domain and execution domain, justify. [02]
- Q - 2 (a) Give brief idea of memory allocation done during compilation with block structured language. What are the limitations of stack based memory allocation? [06]

OR

- (a) In block structured language, what is the scope of a variable and how to access local and non local variable? Explain with suitable example. [06]
- (b) Find Register Requirement (RR) for the below given expression [04]
$$f + (x + y) * ((a + b) / (c - d))$$
- (c) Explain in brief functions of analysis phase and synthesis phase of the assembler. [04]
- Q - 3 (a) What is the significance of intermediate code phase in compiler? What are the ways to represent intermediate code? [06]
- (b) What is Editor? What are the functions of Editor? [04]
- (c) What do you mean by debugging a program? What are the methods available for debugging? [04]

OR

- Q - 3 (a) What is the goal of code optimization? Explain dead code elimination and constant propagation with example. [06]
- (b) Write a short note on "Forms of Text Editors". [04]
- (c) Write a short note on types of Macro Expansion. [04]

SECTION - II

- Q - 4 (a) What is language processor? What are the spectrums of language processor? [05]
- (b) State true or false with justification: "Programs using dynamic memory allocation scheme is executed slowly than static memory allocation". [02]
- Q - 5 (a) Compare Call by value, Call by name and Call by reference method with an example. [06]

OR

- (a) What is grammar? List and explain various types of grammar. [06]
- (b) Explain with an example how forward reference problem is solved in one pass assembler? [04]

- (c) Draw the DFA for the below given regular expression.

[04]

$(0|1)^*01$

- Q - 6 (a) Given the following operator grammar,

[07]

$S \rightarrow | - A - |$

$A \rightarrow VaB \mid ^$

$B \rightarrow VaC$

$C \rightarrow VbA$

$V \rightarrow id$

(1) Construct an operator precedence matrix for the operators of the grammar.

(2) Give a bottom up parse for a string containing three <id> symbols.

OR

- (a) For given grammar,

[07]

$E \rightarrow T + E \mid T - E \mid T$

$T \rightarrow T * V \mid T / V \mid V$

$V \rightarrow a|b|c|d$

Make Top Down Parsing without backtracking for the string $a + b / c - a * d$.

- (b) Consider the following grammar and prove below grammar is ambiguous.

[03]

$S \rightarrow aSbS \mid bSaS \mid ^$

OR

- (b) Remove the left recursion from the following grammar.

[03]

$S \rightarrow Aa \mid b$

$A \rightarrow Ac \mid Aad \mid bd \mid ^$

- (c) What is a loader? What are the functions of loader?

[04]